Finite-Time Swarm Tracking: Numerical Results

All quantities are normalized. This report documents the rotating and compressing multi-agent star formation tracking scenario with transient collision avoidance, retaining the arbitrary formation demonstration (Exp. 1), the finite-time convergence and coupling gain evaluation under rotating star formation (Exp. 2), the certified safe versus unsafe transient comparison across a 2×3 grid (Exp. 3), and the Experiment 4 coupling–compression rate parameter sweep.

1 Model and parameters

The physical swarm consists of $n = 10$ agents in $\mathbb{R}^2$ governed by the closed-loop velocity command:

$$\dot{x}_i = u_i = \dot{q}_i - \delta_i - \mathrm{sig}^{\beta}(\delta_i) - \sum_{j \in \mathcal{N}_i(t)} w_{ij} M(\delta_i - \delta_j), \qquad \delta_i = x_i - q_i,$$

where $\mathrm{sig}^{\beta}(z) = \mathrm{sign}(z)|z|^{\beta}$, $M = mI_2$, $\beta = 0.5$, $d_{\mathrm{safe}} = 1.0$ m, and $R_c = 6.0$ m. The state-dependent interaction weights are

$$w_{ij}(x,t) = \frac{a_{ij}(t)}{\|x_i - x_j\| + 0.2}, \qquad a_{ij}(t) = \begin{cases} 1 & \|x_i - x_j\| < R_c \ (i \neq j), \\ 0 & \text{otherwise}, \end{cases}$$

giving lower bound $\underline{w} = 1/(R_c + 0.2) = 1/6.8$ and algebraic connectivity $\lambda_2(L_{\mathcal{H}}) = 10.0$.

The rotating and compressing target stations $q_i(t)$ follow:

$$r(t) = \begin{bmatrix} t \\ 2\sin(0.5t) \end{bmatrix}, \qquad \dot{r}(t) = \begin{bmatrix} 1 \\ \cos(0.5t) \end{bmatrix},$$
$$q_i(t) = r(t) + a(t)R(\omega t)p_i^0, \qquad \dot{q}_i(t) = \dot{r}(t) + \dot{a}(t)R(\omega t)p_i^0 + a(t)\dot{R}(\omega t)p_i^0,$$

where $p_i^0 = b_i[\cos\phi_i, \sin\phi_i]^T$, $\phi_i = \pi/2 + \pi i/5$, and $b_i = 2.8$ for even i and 1.22 for odd i ($i = 0, \ldots, 9$), centered such that $\sum_i p_i^0 = 0$. The baseline minimum pairwise distance is $DQ_0 \approx 1.4342$ m, and the template diameter is $\ell_{\mathcal{H},0} \approx 5.3259$ m. Rotation operates at angular velocity $\omega = \pi/4$ rad/s (45°/s) via

$$R(\omega t) = \begin{bmatrix} \cos(\omega t) & -\sin(\omega t) \\ \sin(\omega t) & \cos(\omega t) \end{bmatrix}, \qquad \dot{R}(\omega t) = \omega \begin{bmatrix} -\sin(\omega t) & -\cos(\omega t) \\ \cos(\omega t) & -\sin(\omega t) \end{bmatrix}.$$

Formation scale contraction follows

$$a(t) = a_\infty + (a_0 - a_\infty)e^{-\kappa t}, \qquad a_0 = 1.10, \quad a_\infty = 0.75, \quad \dot{a}(t) = -\kappa(a_0 - a_\infty)e^{-\kappa t}.$$

Notice that for all $t \geq 0$, the station plan satisfies $d_q(t) \geq a_\infty DQ_0 = 1.0757$ m $> d_{\mathrm{safe}} = 1.0$ m and $\ell_q(t) \leq a_0 \ell_{\mathcal{H},0} = 5.8585$ m $< R_c = 6.0$ m. The assigned stations are strictly safe and connected for all time.

The initial perturbation $\delta_i(0)$ is zero-mean ($\sum_i \delta_i(0) = 0$), generated via random annular ring perturbation around each target station $q_i(0)$ with radius $r_i \in [0.3, 0.7]$ m ($d_{\mathrm{safe}} = 1.0$ m, $R_c = 6.0$ m, seed 123). At $t = 0$, physical separation is safe ($d_x(0) = 1.0864$ m $> d_{\mathrm{safe}}$) and connected ($\ell_x(0) = 5.7419$ m $< R_c$), with initial error norm $\|\delta_0\| \approx 1.6532$, initial disagreement distortion $\rho_0 = \sqrt{2}\|\eta_0\| \approx 2.3380$ m, and universal settling deadline $\overline{T} = 2\ln(1 + \sqrt{\|\delta_0\|}) \approx 1.6534$ s.

The geometric clearance margin is $s(t) = \min\{d_q(t) - d_{\mathrm{safe}},\ R_c - \ell_{\mathcal{H}}(t)\} = \min\{a(t)DQ_0 - d_{\mathrm{safe}},\ R_c - a(t)\ell_{\mathcal{H},0}\} > 0$. For coupling gain m, the effective decay rate is $\gamma = 1 + m\underline{w}\lambda_2(L_{\mathcal{H}}) = 1 + m\frac{10}{6.8}$, and the required rate is

$$\gamma_{\mathrm{req}}(\kappa) = \sup_{\tau \in (0, \overline{T}]} \frac{\ln(\rho_0 / s(\tau))}{\tau}.$$

The dynamic disagreement distortion tube is bounded by

$$\rho_{\mathcal{H}}(t) = \min\{\sqrt{2}R(t),\ \rho_0 e^{-\gamma t}\}, \qquad R(t) = \max\left\{0, \left((1+\sqrt{\|\delta_0\|})e^{-\frac{1}{2}t} - 1\right)^2\right\},$$

where $R(t)$ is the pure Theorem 1 station tracking bound (independent of graph and coupling gain m).

2 Formation demonstration

The multi-formation reconfiguring demonstration (Exp. 1) uses smooth quintic blends across hexagon, wedge, star, and arrow templates. It demonstrates continuous tracking of reconfiguring formations under switching interaction topologies.

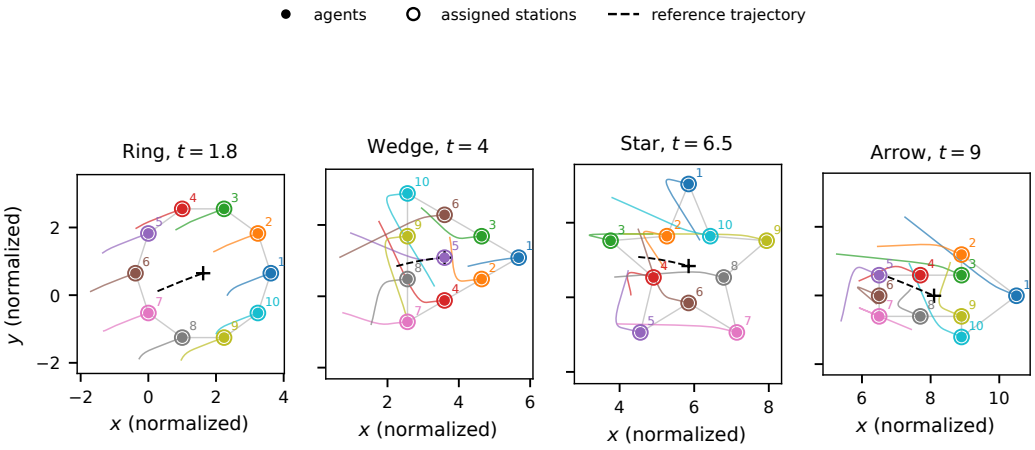


Figure 1: Experiment 1: Moving reference and reconfiguring formations.

3 Finite-time tracking and coupling gain evaluation

Experiment 2 evaluates the rotating-compressing star formation maneuver ($\kappa = 2.0$) across coupling gains $m \in \{0, 0.5, 2, 2.5, 5, 10\}$. Figure 2 depicts station tracking error $\|\delta(t)\|_2$, formation disagreement $\|\eta(t)\|_2$, and the initial control command $\|u(t)\|_2$. In panels (a) and (b), error curves are cut at 10^{-4} to present the clean finite-time transient down to exact settling.

Increasing the coupling gain m accelerates disagreement decay, reducing the settling time to the 10^{-6} error threshold from 1.1350 s at $m = 0$ down to 0.2250 s at $m = 10$ (Table 1). All runs settle well within the theoretical universal deadline $\overline{T} = 1.6534$ s.

Table 1: Experiment 2: Settling times and control metrics across coupling gains m under rotating star formation ($\kappa = 2.0$).

m	$t_{10^{-6}}$ [s]	$\overline{T}$ [s]	Peak $\|u\|_2$	J_u
0.0	1.1350	1.6534	9.7818	389.00
0.5	0.8450	1.6534	11.3082	391.62
2.0	0.5350	1.6534	18.6318	399.15
2.5	0.4933	1.6534	21.4739	401.63
5.0	0.3467	1.6534	36.5758	414.04
10.0	0.2250	1.6534	68.0054	438.80

4 Star formation safety and connectivity transient

Experiment 3 analyzes the fast compression maneuver ($\kappa = 6.0$) over $t \in [0, 2.5]$ s, comparing two regimes in a 2×3 grid format (Figure 3):

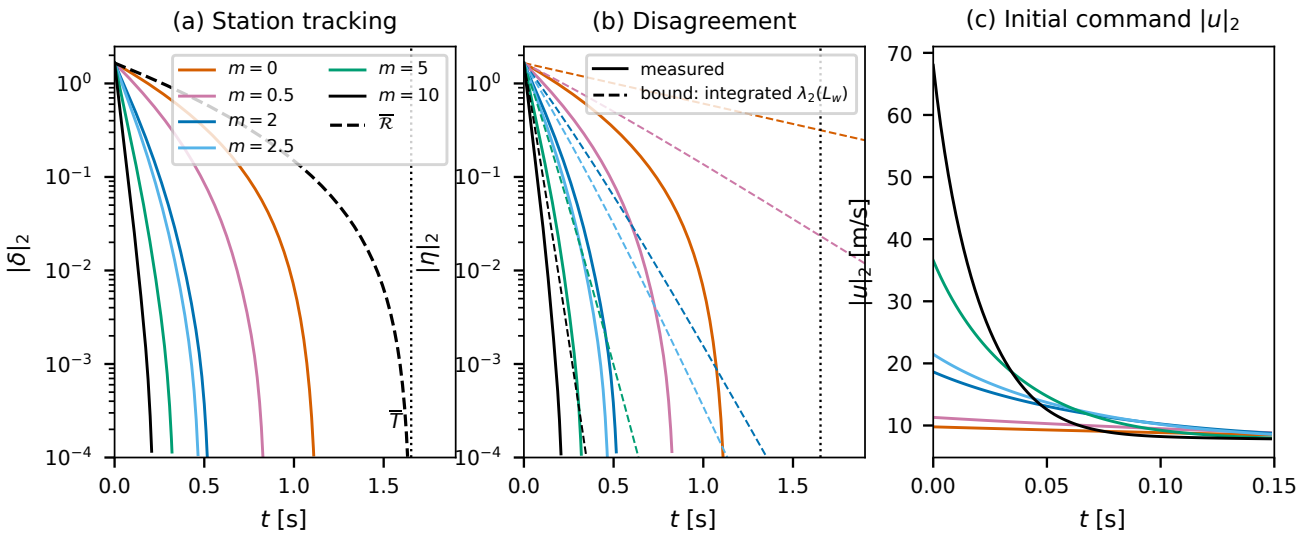


Figure 2: Experiment 2 finite-time tracking and disagreement across coupling gains m ($\kappa = 2.0$). (a) Station tracking $\|\delta\|_2$ against universal bound $R(t)$ and deadline $\overline{T}$, cut at 10^{-4}. (b) Disagreement $\|\eta\|_2$ against exponential rate bound, cut at 10^{-4}. (c) Initial physical control command $\|u\|_2$ ($t \in [0, 0.15]$ s).

- **Row 1 (Safe Case, $m = 2.5$):** With coupling gain $m = 2.5$, the convergence rate is $\gamma = 4.6765$. In simulation, the swarm settles within $t_{\text{set}} = 0.4600$ s. Both minimum pairwise separation ($d_{x,\min} = 1.0756$ m $> d_{\text{safe}} = 1.0$ m) and maximum diameter ($\ell_{x,\max} = 5.7419$ m $< R_c = 6.0$ m) are strictly preserved across the entire transient without violation.
- **Row 2 (Unsafe Mid-Transient Collision, $m = 0.0$):** With uncoupled agents ($m = 0.0$), the decay rate is only $\gamma = 1.0000$. Although stations remain strictly safe ($d_q(t) \geq 1.0757$ m), rapid maneuver compression outpaces uncoupled error convergence. As a result, the uncoupled swarm suffers a genuine physical collision mid-transient: $d_x(t) < d_{\text{safe}}$ for $t \in [0.11, 0.50]$ s (duration 0.39 s), reaching a minimum distance of $d_{x,\min} = 0.9548$ m (a collision breach of 4.52 cm). Network connectivity is maintained ($\ell_{x,\max} = 5.7419$ m $< R_c$).

Table 2: Experiment 3 outcomes: Safe ($m = 2.5$) vs. uncertified mid-transient collision ($m = 0.0$) under fast compression ($\kappa = 6.0$, $t \in [0, 2.5]$ s).

Case	m	κ	γ	γ_{req}	t_{set} [s]	$d_{x,\min}$	$\ell_{x,\max}$	Peak $\|u\|$	J_u	Outcome
Safe	2.5	6.0	4.6765	–	0.4600	1.0756	5.7419	23.2593	402.93	Uncertified; Safe
Unsafe	0.0	6.0	1.0000	–	1.1317	0.9548	5.7419	16.1726	393.53	Uncertified; Safety Violation

Figure 3 presents the 2×3 transient response comparison. Column 1 displays the minimum pairwise distance $d_x(t)$ against the lower tube bound $d_q(t) - \rho_{\mathcal{H}}(t)$ and threshold $d_{\text{safe}} = 1.0$ m, with the collision breach shaded in red for Row 2. Column 2 shows maximum pairwise distance $\ell_x(t)$ against $\ell_q(t) + \rho_{\mathcal{H}}(t)$ and connectivity limit $R_c = 6.0$ m. Column 3 compares geometric margin $s(t)$ against distortion tube radius $\rho_{\mathcal{H}}(t)$, highlighting certified ($s \geq \rho_{\mathcal{H}}$) in green and uncertified ($s < \rho_{\mathcal{H}}$) in orange.

5 Coupling–speed parameter map

Experiment 4 maps the parameter space over a 4×4 grid: coupling gain $m \in \{0.0, 1.5, 3.0, 5.0\}$ and maneuver compression rate $\kappa \in \{2.0, 4.0, 6.0, 8.0\}$. For each parameter pair, the sufficient condition $1 + m\underline{w}\lambda_2(L_{\mathcal{H}}) > \gamma_{\text{req}}(\kappa)$ is evaluated against simulated physical separation and connectivity.

The grid yields 14 uncertified but observed safe configurations (Category B) and 2 observed collision cases (Category C at $m = 0.0, \kappa \geq 6.0$, where minimum distance drops to 0.909082 m < 1.0 m). All coupled configurations ($m \geq 1.5$) preserve both safety ($d_{x,\min} \geq 1.0756$ m) and connectivity ($\ell_{x,\max} = 5.7419$ m) across all compression speeds without any validation failure. Peak control effort spans 9.78177–37.378.

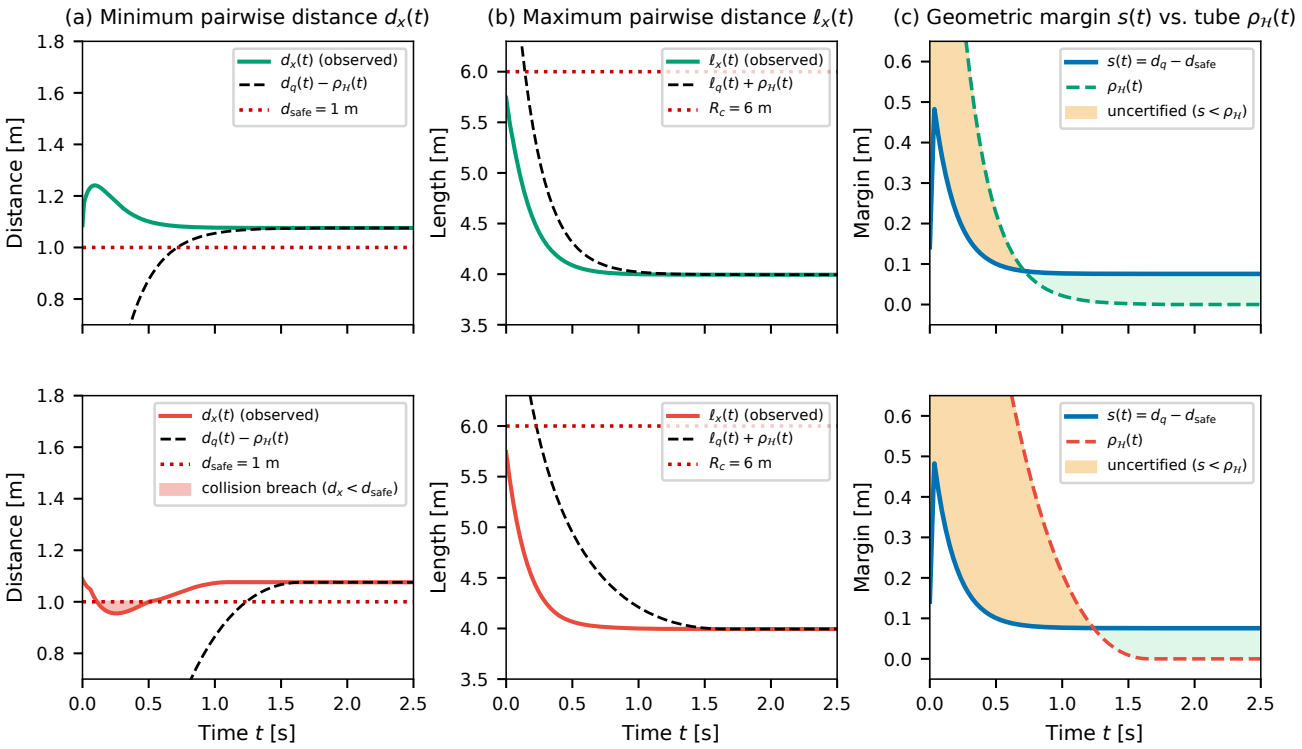


Figure 3: Experiment 3: Safety and connectivity during fast star compression ($\kappa = 6.0$, $t \in [0, 2.5]$ s). Top row: Safe ($m = 2.5$). Bottom row: Unsafe Mid-Transient Collision ($m = 0.0$). Columns show minimum pairwise distance $d_x(t)$ (left), maximum pairwise distance $\ell_x(t)$ (middle), and geometric margin $s(t)$ vs. distortion tube $\rho_{\mathcal{H}}(t)$ (right).

Figure 4 (left) overlays the continuous sufficient gain boundary $m_{\text{req}}(\kappa) = \max\left(0, \frac{\gamma_{\text{req}}(\kappa)-1}{\underline{w}\lambda_2(L_{\mathcal{H}})}\right)$ across $\kappa \in [2.0, 8.0]$. While uncoupled agents ($m = 0$) collide under aggressive compression ($\kappa \geq 6.0$), introducing positive coupling gain $m \geq 1.5$ completely eliminates transient collisions. Figure 4 (right) presents the corresponding peak control command map.

6 Reproduction and limits

All experiments can be reproduced using:

- Experiment 2: `python3 run_experiments.py --experiment 2`
- Experiment 3: `python3 run_experiments.py --experiment 3`
- Experiment 4: `python3 run_experiments.py --experiment 4 --workers 3`

Compile the document from `ACC_paper/` using `pdflatex numerical_results.tex`.

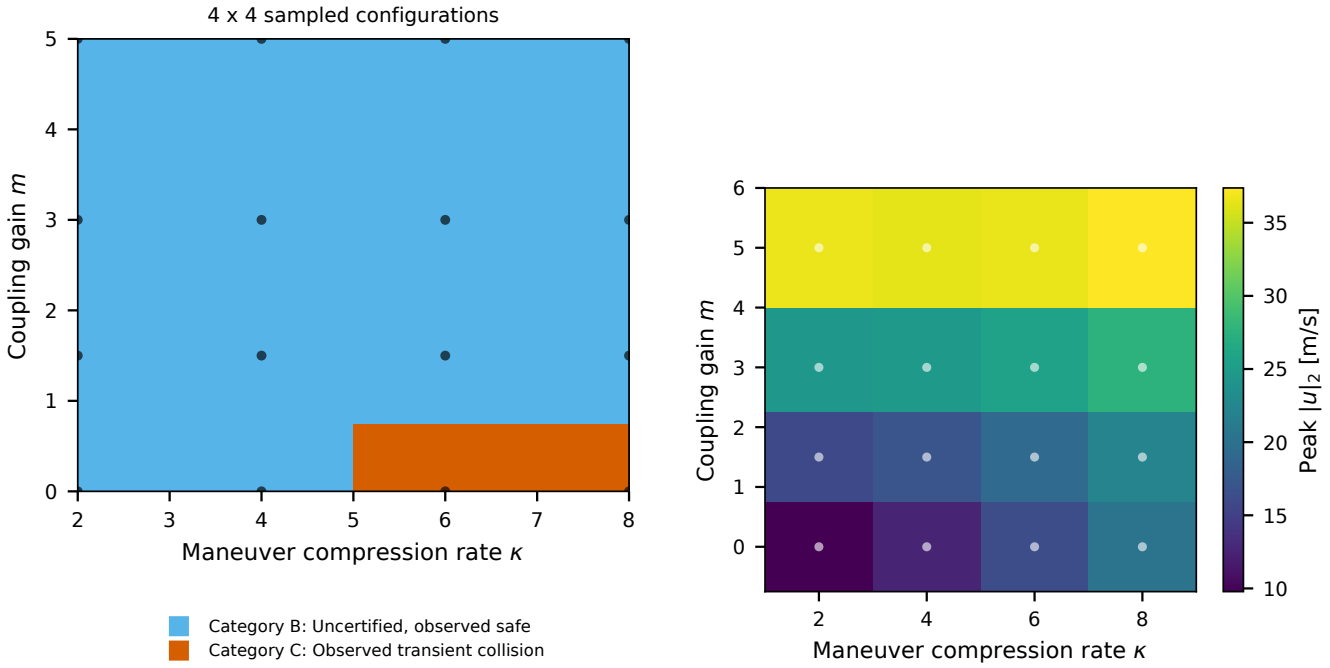


Figure 4: Experiment 4 parameter sweep over $m \in \{0.0, 1.5, 3.0, 5.0\}$ and $\kappa \in \{2.0, 4.0, 6.0, 8.0\}$. Left: Safety classification map with continuous theoretical design boundary $m_{\text{req}}(\kappa)$. Right: Peak control command $\|u\|_{\max}$.